Is *not* prefix palindromic length of a k-automatic word k-regular?

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Prefix palindromic length

The palindromic length $PL(u)$ is the minimal number of concatenated palindromes needed to express the word u.

$$abbaba = (abba)\ (b)\ (a) = (a)\ (bb)\ (aba)$$

$$PL(abbaba) = 3$$

The prefix palindromic length $PPL_{\mathbf{u}}(n)$ of an infinite word $\mathbf{u}$ is the palindromic length of the prefix of length n of $\mathbf{u}$.

Prefix palindromic length

$$\mathbf{t} = abbabaabba\cdots$$

n	1	2	3	4	5	6	7	8	9	10
$PPL_{\mathbf{t}}(n)$	1	2	2	1	2	3	3	2	3	4

What is known on the prefix palindromic length?

Very little.

If $\mathbf{u}$ is periodic, it can be bounded

$$\mathbf{u} = ababababab\cdots$$

or not

$$\mathbf{u} = abcabcabcabc\cdots .$$

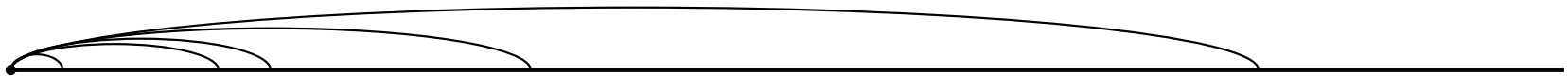
Conjecture

Conjecture (F., Puzynina, Zamboni, 2013)

For every non ultimately periodic word $\mathbf{u}$, the function $PPL_{\mathbf{u}}(n)$ is unbounded.

Proven for u avoiding some power k, that is, words $v^k = \underbrace{v \cdots v}_{k}$ (2013).

The proof is not constructive.
The number of palindromes starting at the same point is logarithmical.



Conjecture

Conjecture (F., Puzynina, Zamboni, 2013)

For every non ultimately periodic word $\mathbf{u}$, the function $PPL_{\mathbf{u}}(n)$ is unbounded.

The conjecture is also proven

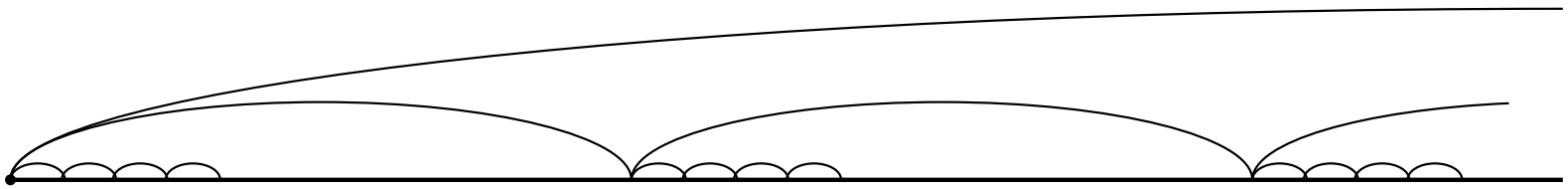
- For all morphic words ([FPZ 2013] + [Klouda, Starosta, 2015])
- In the greedy versions [Bucci, Richomme 2018]
- For all Sturmian words [F. 2018]

Conjecture

Conjecture (F., Puzynina, Zamboni, 2013)

For every non ultimately periodic word $\mathbf{u}$, the function $PPL_{\mathbf{u}}(n)$ is unbounded.

So, the conjecture holds for *almost all* infinite words, and here are the remaining cases.



What about known words?

It is easier to find a decomposition to few palindromes than to prove it does not exist.

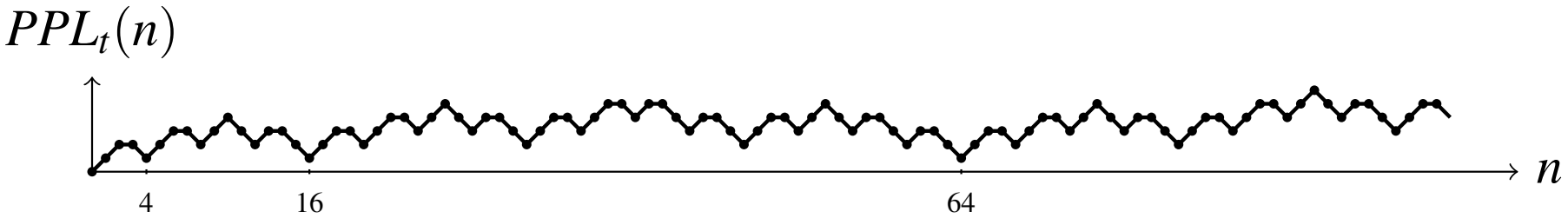
Upper bounds :
F. 2017 - Fibonacci word;
Ambrož, Kadlec, Masáková, Pelantová, 2019: morphic words from class $\mathcal{P}$

Lower bounds: F. 2019; Shuo LI 2020 - for some Toeplitz words including the period doubling word.

Thue-Morse word

$$\sigma : \begin{cases} a \to abba, \\ b \to baab. \end{cases}$$

$$\mathbf{t} = \sigma(\mathbf{t}) = abba\ baab\ baab\ abba\ baab\ abba \cdots$$



Thue-Morse word

Theorem (F., 2019)

The following identities hold for all $n \geq 0$:

$$\begin{aligned} PPL_{\mathbf{t}}(4n) &= PPL_{\mathbf{t}}(n), \\ PPL_{\mathbf{t}}(4n+1) &= PPL_{\mathbf{t}}t(n) + 1, \\ PPL_{\mathbf{t}}(4n+2) &= \min(PPL_{\mathbf{t}}(n), PPL_{\mathbf{t}}(n+1)) + 2, \\ PPL_{\mathbf{t}}(4n+3) &= PPL_{\mathbf{t}}(n+1) + 1. \end{aligned}$$

Formula rediscovered by Shuo LI later the same year.

Morphism for first differences

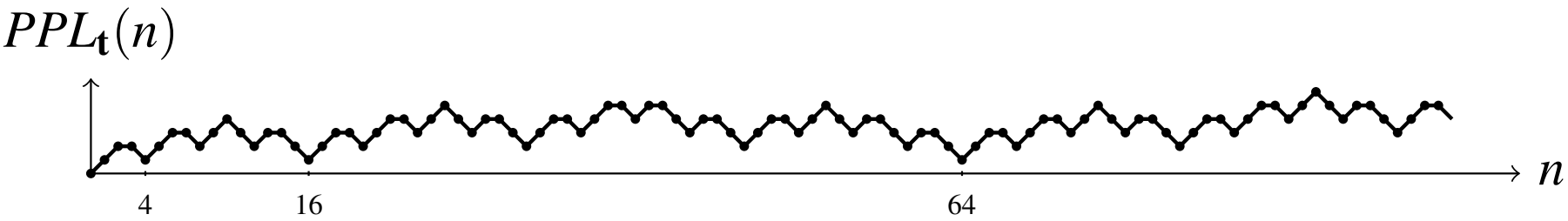
$$d_{\mathbf{t}}(n) = PPL_{\mathbf{t}}(n+1) - PPL_{\mathbf{t}}(n) \in \{\diagdown, \text{—}, \diagup\}$$

Proposition

The sequence $(d_{\mathbf{t}}(n))$ is the fixed point of the morphism

$$\delta : \begin{cases} \diagup \to \diagup\diagdown \\ \text{—} \to \diagup\diagdown \\ \diagdown \to \diagup\diagdown \end{cases}$$

Growth



$$\limsup \frac{PPL_{\mathbf{t}}(n)}{\ln n} = \frac{3}{4\ln 2}.$$

k-regularity?

The Thue-Morse word $\mathbf{t}$ is 2-automatic (and 4-automatic).

Its sequence $PPL_{\mathbf{t}}(n)$ is 4-regular (and 2-regular).

The first differences $d_{\mathbf{t}}(n)$ are 2- and 4-automatic.

Question

Is $PPL_{\mathbf{u}}(n)$ a k-regular sequence whenever $\mathbf{u}$ is k-automatic?

k-automatic words

Definition

A word $\mathbf{u} = \mathbf{u}[0] \cdots \mathbf{u}[n] \cdots$ is *k-automatic* if there exists a DFAO A such that every symbol $\mathbf{u}[n]$ of $\mathbf{u}$ can be obtained as the output of A with the base-k representation of n as the input.

Morphic definition

A word $\mathbf{u}$ is k-automatic if and only if $\mathbf{u} = \psi(\varphi^\omega(a))$ for some k-uniform morphism φ and a uniform morphism ψ.

Kernel definition

A word $\mathbf{u}$ is k-automatic if and only if its k-kernel

$$\ker_k(\mathbf{u}) = \{(\mathbf{u}[k^e n + b])_{n \geq 0} : e \geq 0, 0 \leq b < k^e\}.$$

is finite.

Equivalence

The three definitions above are equivalent.

k-regularity

Definition

A $\mathbb{Z}$-valued sequence is *k-regular* if the $\mathbb{Z}$-module generated by its k-kernel is finitely generated.

Charlier, Rampersad, Shallit, 2012

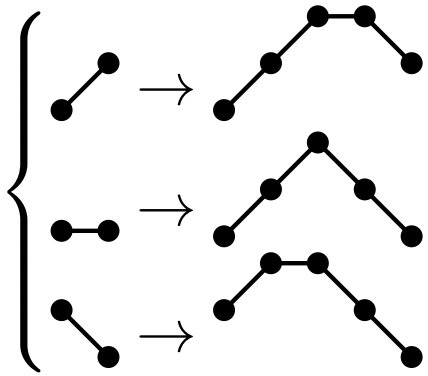
Many functions of k-automatic words are k-regular, including

- factor complexity,
- number of different palindrome factors of length n.

Walnut software

What about our case?

- That method does not work for the PPL function.
- The situation looks complicated even for simplest examples like the period-doubling word.
- On the good side: $PPL_{\mathbf{u}}$ is k-regular if and only if $d_{\mathbf{u}}$ is k-automatic.

δ : 

or, equivalently, $\begin{cases} + \mapsto ++0-, \\ 0 \mapsto ++--, \\ - \mapsto +0--. \end{cases}$

First differences

Proposition (Particular case of a lemma by Saarela, 2017)

For every infinite word $\mathbf{u}$, the first differences of the prefix palindromic length is a sequence over the alphabet $\{-1, 0, 1\}$ which we prefer to denote as $\{-, 0, +\}$.

So, it is a word over a finite alphabet which is k-automatic if and only if $PPL_{\mathbf{u}}$ is k-regular.

The best result we get

Theorem

If a k-automatic word $\mathbf{u}$ contains a finite number of distinct palindromes, then the PPL-difference sequence $d_{\mathbf{u}}$ is k-automatic.

IDEA OF THE PROOF: If p is the length of a longest palindrome in $\mathbf{u}$, we have

$$PPL(n) = \min\{PPL(n-k)+1 : k = 1, \ldots, p, \mathbf{u}[n-k+1..n] \text{ is a palindrome}\}.$$

Just a finite number of cases to check and a transducer from *another* theorem by Cobham (1972).

Paperfolding word

$$\mathbf{u}_{pf} = \psi(\varphi_{pf}^{\omega}(a)) = 0010011000110110\cdots,$$

where

$$\varphi_{pf}: \begin{cases} a \mapsto ab, \\ b \mapsto cb, \\ c \mapsto ad, \\ d \mapsto cd, \end{cases}$$

and the coding ψ is defined as $\psi(a) = \psi(b) = 0$, $\psi(c) = \psi(d) = 1$.

PPL of the paperfolding word

The longest palindromes in the paperfolding word are of length 13.

Theorem

The sequence d_{pf} over the alphabet $\{-, 0, +\}$ is equal to $d_{pf} = \gamma_{pf}(\mu_{pf}^{\omega}(a_0))$, where

$$\mu_{pf}: \begin{cases} a_0 \mapsto a_0 b_a, & & & \\ a_b \mapsto a_b b_a, & b_a \mapsto c_b b_c, & c_b \mapsto a_b d_a, & d_a \mapsto c_b d_c, \\ a_d \mapsto a_d b_a, & b_c \mapsto c_d b_c, & c_d \mapsto a_d d_a, & d_c \mapsto c_d d_c \end{cases}$$

and

PPL of the paperfolding word

$$\gamma_{pf}: \begin{cases} a_0 \mapsto +0+0-+0+000-++0-+-P00+00+0-+000+000+00000-0++0-+0-, \\ a_b \mapsto 0++-0+00+00-0+0000P00+00+0-+000+000+00000-0++0-+0-, \\ a_d \mapsto 0+00000+00000+000-P00+00+0-+000+000+00000-0++0-+0-, \\ b_a \mapsto 0++-0+00+00-0+0000P+-+0-0+000+000+0+-+0-000+000+0-, \\ b_c \mapsto +00+-00+0000+0000-P+-+0-0+000+000+0+-+0-000+000+0-, \\ c_b \mapsto 0++-0+00+00-0+0000P00+00+0-+000+000+00000-0++0-+00, \\ c_d \mapsto 0+00000+00000+000-P00+00+0-+000+000+00000-0++0-+00, \\ d_a \mapsto 0++-0+00+00-0+0000P+-+0-0+000+000+0+-+0-000+000+00, \\ d_c \mapsto +00+-00+0000+0000-P+-+0-0+000+000+0+-+0-000+000+00 \end{cases}$$

with $P = 0+00+00-0++-0+0$.

Rudin-Shapiro word

The Rudin-Shapiro word $\mathbf{u}_{rs}$ is the 2-automatic word

$$\mathbf{u}_{rs} = \psi(\varphi_{rs}^{\omega}(a)) = 00010010000111010\cdots,$$

where

$$\varphi_{rs}: \begin{cases} a \to ab, \\ b \to ac, \\ c \to db, \\ d \to dc, \end{cases}$$

and the coding ψ is defined by $\psi(a) = \psi(b) = 0$, $\psi(c) = \psi(d) = 1$.

The longest palindromes in the Rudin-Shapiro word are of length 14.

PPL of the Rudin-Shapiro word

Theorem

The sequence d_{rs} over the alphabet $\{-, 0, +\}$ is equal to

$$d_{rs} = \gamma_{rs}(\mu_{rs}^{\omega}(A)),$$

where

$$\mu_{rs}: \begin{cases} A \to AB, \\ B \to CD, \\ C \to EB, \\ D \to ED, \\ E \to CB \end{cases}$$

and

PPL of the Rudin-Shapiro word

$$\gamma_{rs}: \begin{cases} A \mapsto +00+00000-++00-++00-+0+00+00+00+-0+00-+00+-0+0+0-0+0P, \\ B \mapsto 0+0-0++-00+0+0-++00-+0+00+00+00+0+0-0++-00+0+0-+000+P, \\ C \mapsto -0+00-+00+-0+0+00+00-++-+00+000+0-+000+-+0-0+0+0-0+0P, \\ D \mapsto -0+00-+00+-0+0+00+00-++-+00+000+0+0-0++-00+0+0-+000+P, \\ E \mapsto 0+0-0++-00+0+0-++00-+0+00+00+00+-0+00-+00+-0+0+0-0+0P. \end{cases}$$

with $P = 0-+00+00+00+.$

Remarks

- The results on the paperfolding word and on the Rudin-Shapiro word are not direct corollaries of the general theorem.
- The Thue-Morse word contains an infinite number of palindromes but its $d_{\mathbf{t}}$ is also 2-automatic.
- We also have a conjecture on the Sierpiński word. ($0 \to 010, 1 \to 111$)

Period-doubling word

$$\mathbf{u}_{pd} = \varphi_{pd}^{\omega}(a) = abaababababaaabaa\cdots,$$

where

$$\varphi_{pd}: \begin{cases} a \to ab, \\ b \to aa. \end{cases}$$

The growth of $\sup PPL$ is logarithmic (Ambrož et al., Li)

2-kernel?

$$\ker_2(\mathbf{u}) = \{\mathbf{u}[b]\mathbf{u}[2^e + b]\mathbf{u}[2.2^e + b]\cdots\mathbf{u}[2^e n + b]\cdots : e \geq 0, 0 \leq b < 2^e\}.$$

u is 2-automatic if and only if its 2-kernel is finite.

Let k_n be the number of elements of the 2-kernel observed in the prefix of $d_{pd}(n)$ pf length 4^n. Then

4^m	4^4	4^5	4^6	4^7	4^8	4^9	4^{10}	$4^{11} = 4194304$
k_m	66	145	297	584	1046	1816	3047	5051
k_m/k_{m-1}	3.0	2.197	2.048	1.966	1.791	1.736	1.678	1.658

Fibonacci word

$\mathbf{u}_f = abaababaabaab\cdots$ is the fixed point of

$$\varphi_f : \begin{cases} a \to ab, \\ b \to a. \end{cases}$$

It is *Fibonacci-automatic*, and all the regularity results can be generalized to it if we use the Fibonacci numeration system (Du, Mousavi, Schaeffer, Shallit, 2016).

Fibonacci-kernel of $d_{\mathbf{f}}(n)$?

$\lvert\varphi_f^{3m}(a)\rvert$	5	21	89	377	1597	6765	28657	121393
k_m	3	11	31	88	207	504	1139	2377
k_m/k_{m-1}		3.67	2.82	2.85	2.35	2.43	2.26	2.09

Main conjecture(s)

Conjecture

The PPL-difference sequence of the period-doubling word ($a \to ab, b \to aa$) is **NOT** 2-automatic.

Conjecture

The PPL-difference sequence of the Fibonacci word ($a \to ab, b \to a$) is **NOT** Fibonacci-automatic.

Conjecture

The PPL-difference sequence of a k-automatic word containing an infinite number of palindromes is **NOT** obliged to be k-automatic.